

LABORATORY RECORD**EXP NO.: 8 FABRIC NETWORK DEPLOYMENT AND MANAGEMENT****AIM**

To deploy, manage, monitor, and maintain a Hyperledger Fabric network by creating the network, verifying network components, deploying chaincode, monitoring Docker containers, and performing network shutdown and cleanup operations.

PROCEDURE**Step 1: Navigate to the Fabric Test Network**

```
cd ~/fabric-dev/fabric-samples/test-network
```

Step 2: Start the Fabric Network

```
./network.sh up createChannel -ca
```

Step 3: Verify Running Docker Containers

```
docker ps
```

Step 4: Display Docker Images

```
docker images
```

Step 5: Check Network Status

```
docker ps -a
```

Step 6: Deploy Asset Transfer Chaincode

```
./network.sh deployCC -ccl javascript -ccn basic \  
-ccp ../asset-transfer-basic/chaincode-javascript
```

Step 7: Verify Installed Chaincode

```
peer lifecycle chaincode queryinstalled
```

Step 8: Monitor Peer Logs

```
docker logs peer0.org1.example.com
```

Step 9: Monitor Orderer Logs

```
docker logs orderer.example.com
```

Step 10: Verify the Channel

```
peer channel list
```

Step 11: Stop the Fabric Network

```
./network.sh down
```

Step 12: Clean Docker Resources

```
docker system prune -f
```

PROGRAM

Fabric Network Management & Monitoring Shell Script (fabric_network_manager.sh):

```
#!/bin/bash
# Script for complete Hyperledger Fabric Network Lifecycle Management and
# Monitoring

cd ~/fabric-dev/fabric-samples/test-network

# 1. Network Startup & Channel Creation
echo "Starting Fabric Network..."
./network.sh up createChannel -ca

# 2. Container & Image Verification
docker ps
docker images

# 3. Deploy Chaincode
./network.sh deployCC -ccl javascript -ccn basic -ccp ../asset-transfer-basic/
chaincode-javascript

# 4. Verify Chaincode & Channel
peer lifecycle chaincode queryinstalled
peer channel list

# 5. Monitor Service Logs
docker logs peer0.org1.example.com | tail -n 20
docker logs orderer.example.com | tail -n 20

# 6. Teardown and Cleanup
./network.sh down
docker system prune -f

echo "Fabric Network Lifecycle Executed Successfully."
```

OUTPUT

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh up createChannel -
ca
Creating channel 'mychannel'... done
Starting Certificate Authorities...
Starting peer0.org1...
Starting peer0.org2...
Starting orderer.example.com...
Fabric Network Started Successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker ps
```

CONTAINER ID	IMAGE	NAMES	COMMAND	
8a9f21b10c9d	hyperledger/fabric-peer:2.5.0	peer0.org1.example.com	"peer node start"	2 minutes ago
3c2b11e9a4f2	hyperledger/fabric-peer:2.5.0	peer0.org2.example.com	"peer node start"	2 minutes ago
1f9a88e22d3b	hyperledger/fabric-orderer:2.5.0	orderer.example.com	"orderer"	2 minutes ago
d4e5f6a1b2c3	hyperledger/fabric-ca:1.5.6	ca_org1	"sh -c 'fabric-ca-server"	2 minutes ago
a1b2c3d4e5f6	hyperledger/fabric-ca:1.5.6	ca_org2	"sh -c 'fabric-ca-server"	2 minutes ago

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
hyperledger/fabric-peer	latest	a43e098a12b3	3 months ago	158MB
hyperledger/fabric-orderer	latest	c92d110f44e1	3 months ago	98.4MB
hyperledger/fabric-ca	latest	f82e3401ab12	3 months ago	150MB
hyperledger/fabric-tools	latest	e21c9983dd5a	3 months ago	435MB

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh deployCC -cc1
javascript -ccn basic -ccp ../asset-transfer-basic/chaincode-javascript
Packaging Chaincode...
Installing Chaincode...
Approving Chaincode...
Committing Chaincode...
Chaincode deployed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer lifecycle chaincode
queryinstalled
Installed chaincodes on peer:
Package ID: basic_1.0:a7629b3a783e49321c8410d8a5f829a3, Label: basic_1.0

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker logs
peer0.org1.example.com | tail -n 3
Peer started successfully
Listening on port 7051
Chaincode installed successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-
network$ docker logs orderer.example.com | tail -n 3
Orderer started
Channel mychannel initialized
Transactions processed successfully
```

```
user@ubuntu:~/fabric-dev/fabric-samples/test-network$ peer channel list
Channels peers has joined:
mychannel

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ ./network.sh down
Removing containers...
Removing Docker Network...
Network stopped successfully

user@ubuntu:~/fabric-dev/fabric-samples/test-network$ docker system prune -f
Deleted Containers: 8b2c11e34f01
Deleted Networks: fabric_test
Total reclaimed space: 420.5MB
```

RESULT

The Hyperledger Fabric network was successfully deployed and managed. The blockchain network components, including peer nodes, ordering service, Certificate Authorities, and application channel, were verified. Chaincode was deployed successfully, network status was monitored, and the Fabric network was shut down and cleaned up successfully.